

	the left). These forces are the horizontal component of the forces at the hinge.
2(a)	Hooke's Law states that the extension (or compression) of a spring is directly proportional to the force required to extend (or compress) it, provided the limit of proportionality is not exceeded. <u>Common mistake:</u> • Used the phrase "Resultant force" • Used the word "displacement"
2(b)	Elastic potential energy stored = $\frac{1}{2} \times 6.8 \times 85 \times 10^{-3}$ = 0.289 J ≈ 0.29 J

2(c)	All the elastic potential energy stored in the spring is converted into kinetic energy of the ball. This kinetic energy is converted to gravitational potential energy and kinetic energy of the ball at point P. $\Rightarrow 0.289 = 42 \times 10^{-3} \times 9.81 \times 0.40 + \frac{1}{2} \times 42 \times 10^{-3} \times v^2$ $\Rightarrow v = 2.43 \text{ m s}^{-1}$ $\approx 2.4 \text{ m s}^{-1}$
2(d)(i)	At point P, the net force acting on the ball is weight W and the normal contact force N. $\Rightarrow W + N = \frac{mv^2}{R}$ When the ball just completes the vertical circle without falling off the track, it means that at that instant, the normal contact force N just becomes equals to zero and the weight provides the force needed for the ball to move in circular motion. $\Rightarrow mg = \frac{mv_{\min}^2}{R}$ $v_{\min}^2 = Rg$ $= 0.20 \times 9.81$ $v_{\min} = 1.4 \text{ m s}^{-1}$
2(d)(ii)	The minimum speed remains unchanged. The ball's acceleration remains as 9.81 m/s^2 and the expression $v_{\min} = \sqrt{Rg}$ is independent of mass.
2(e)	Diffraction is the bending or spreading of waves when they travel through a small